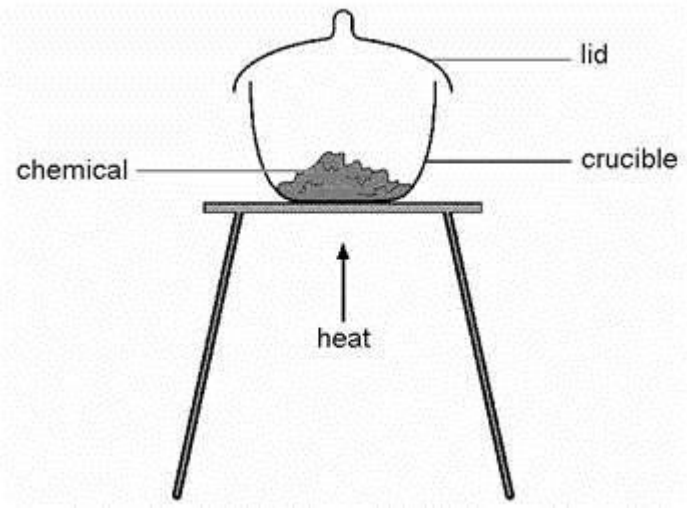
Q1.

Shuli investigated differences between physical and chemical changes.

She put three chemicals in separate crucibles and weighed each one.
She heated each crucible as shown below.
She weighed each crucible again when it had cooled down.



She recorded her observations in a table as shown below.

experiment	name of chemical	observations	change in mass
A	magnesium (a silvery solid)	The silvery magnesium burned brightly in air. A white powder was formed.	increase
B	potassium permanganate (purple crystals)	The purple crystals crackled and turned black. A colourless gas was given off.	decrease
C	zinc oxide (a white powder)	The white powder turned pale yellow on heating. It turned white again on cooling.	no change

(a) (i) In experiment A, magnesium reacts with a gas in the air.

Complete the word equation for the reaction in experiment A.

magnesium + ... → .....

2 marks

(ii) Explain the increase in mass in experiment A. Use your word equation to help you.

.....

.....

1 mark

(b) The gas given off in experiment B re-lit a glowing splint.
Give the name of this gas.

.....

1 mark

(c) Name the white powder left at the end of experiment C.

.....

1 mark

(d) In each experiment, did a chemical change or a physical change take place?
Tick **one** box for each experiment.

experiment	chemical change	physical change
A		
B		
C		

1 mark
Maximum 6 marks

Q1.

(a) (i) *magnesium* + oxygen → magnesium oxide

*do **not** accept formulae*

2 (L6)

(ii) any **one** from

*do **not** accept 'air' for oxygen*

- the oxygen had mass
- oxygen was added to the magnesium

  accept 'magnesium has gained an element'
- the magnesium has reacted with oxygen

  accept 'magnesium is now part of a compound'

1 (L6)

(b) oxygen

accept 'O_2'

1 (L5)

(c) zinc oxide

accept 'ZnO'

1 (L6)

(d)

	chemical change	physical change
A	✓	
B	✓	
C		✓

***all three** ticks are required for the mark*

1 (L6)

[6]